

Shreejit Verma Problem Set 2

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7:27 AM

Q1 Assume that $f: [0, 3] \rightarrow \mathbb{R}$ is integrable and satisfies

$$f(x+1) = \frac{1}{2} f(x) \text{ for all } x \in [0, 2]$$

$$\frac{f(x+1) - f(x)}{1} = -\frac{1}{2} f(x) \quad f(3) = \frac{1}{2} f(1) \quad f(3) - f(1) = -\frac{1}{2} f(1)$$

$$\Rightarrow f'(x) = -\frac{1}{2} f(x) \quad \Rightarrow f(1) - f(0) = -\frac{1}{2}$$

$$\Rightarrow f(x) = -2 f'(x)$$

$$\int_0^3 f(x) dx = \int_0^1 f(x) dx + \int_1^3 f(x) dx$$

$$= 1 + 2 \int_1^3 f'(x) dx$$

$$= 1 + (-2) [f(3) - f(1)] = 1 + f(1)$$

Q2 Assume that f is differentiable on $[0, 1]$ and satisfies

$$f(0) = \frac{1}{2}, f(1) = 1, \text{ and}$$

$$f'(x) = (f(x))^2 \text{ for all } x \in [0, 1]$$

Evaluate $\int_0^1 f(x)^4 dx$

$$\int_0^1 (f(x)^2)^2 dx \quad \Rightarrow \quad f(x) + C = \int (f(x))^2 dx$$

Q3 For each improper integral, determine whether or not it converges.

a) $\int_0^\infty e^{-x^4} dx$

Soln $\int_0^\infty e^{-x^4} dx = -\frac{\Gamma(\frac{1}{4}, x^4) x}{4|x|}$

$$= \frac{\Gamma(\frac{1}{4})}{4} \Rightarrow \text{converges.}$$

b) $\int_0^\infty \frac{1+e^{-x}}{x} dx$

Soln $\int_0^\infty \frac{1+e^{-x}}{x} dx = \int_0^\infty \frac{e^{-x}(e^x+1)}{x} dx$

$$= \int_0^\infty \left(\frac{e^{-x}}{x} + \frac{1}{x} \right) dx$$

$$= \int_0^\infty \frac{e^{-x}}{x} dx + \int_0^\infty \frac{1}{x} dx$$

$$= \left(-\int_0^\infty \frac{e^{-x}}{x} dx \right) + [\ln x]_0^\infty$$

It makes the integral divergent.

c) $\int_1^\infty \frac{x^2}{\sqrt{1+x^6}} dx$

Soln $\int_1^\infty \frac{x^2}{\sqrt{1+x^6}} dx = \int_1^\infty \frac{x^x}{\sqrt{1+t^2}} \frac{dt}{3x^2} \quad t=x^3 \quad \frac{dt}{dx} = 3x^2$

$$= \frac{1}{3} \int_1^\infty \frac{1}{\sqrt{1+t^2}} dt \quad \Rightarrow du = \frac{dt}{3x^2}$$

$$= \frac{1}{3} \int_1^\infty \frac{\sec^2 u du}{\sqrt{1+\tan^2 u}} \quad t = \tan(u) \quad dt = \sec^2(u) du$$

$$= \frac{1}{3} \int_1^\infty \frac{\sec^2 u}{\sec u} du \quad (\because 1+\tan^2 u = \sec^2 u)$$

$$= \frac{1}{3} \int_1^\infty \sec u du$$

$$= \frac{1}{3} \int_1^\infty \frac{\sec u (\tan u + \sec u)}{\tan u + \sec u} du$$

$$= \frac{1}{3} \int_1^\infty \frac{\sec u \tan u + \sec^2 u}{\tan u + \sec u} du$$

$$= \frac{1}{3} \int_1^\infty \frac{1}{v} dv \quad v = \sec u + \tan u$$

$$= \frac{1}{3} \int_1^\infty \ln v$$

$$= \frac{1}{3} [\ln(\tan u + \sec u)]_1^\infty$$

$$= \frac{1}{3} [\ln(\sqrt{t^2+1} + t)]_1^\infty$$

$$= \frac{1}{3} [\ln(\sqrt{x^6+1} + x^3)]_1^\infty$$

This doesn't converge.

Q4 Find all values of p for which the improper integral below converges and evaluate the integral for those values of p .

$$\int_e^\infty \frac{1}{x(\ln x)^p} dx$$

Soln $\int_e^\infty \frac{1}{x(\ln x)^p} dx = \int_e^\infty \frac{1}{x} \frac{1}{t^p} x dt \quad t = \ln x \quad \frac{dt}{dx} = \frac{1}{x}$

$$= \int_e^\infty \frac{1}{t^p} dt \quad \Rightarrow dx = x dt$$

$$= \left[\frac{t^{1-p}}{1-p} \right]_e^\infty$$

$$= \left[\frac{\ln x}{1-p} \right]_e^\infty \quad \Rightarrow p \neq 1$$

$$= \frac{1}{1-p} \left[\ln \infty^{1-p} - \ln e^{1-p} \right]$$

$$= \frac{1}{1-p} [0 - 1] = \frac{1}{p-1} \quad \left. \begin{array}{l} p > 0 \\ p \neq 1 \end{array} \right\}$$

Q5 Find all values of $p \in \mathbb{R}$ for which the improper integral converges

$$\int_e^\infty \frac{(\ln x)^p}{x^2} dx$$

Soln $\int_e^\infty \frac{(\ln x)^p}{x^2} dx = \int \frac{t^p}{x^2} x dt \quad t = \ln x \quad \text{Also } \frac{1}{x} = e^{-t}$

$$= \int \frac{t^p}{x} dt \quad \frac{dt}{dx} = \frac{1}{x}$$

$$= \int t^p e^{-t} dt \quad \Rightarrow dx = x dt$$

$$= \frac{1}{p+1} \int e^{-u^{\frac{1}{p+1}}} du \quad u = t^{p+1}$$

This is type of gamma function $\frac{du}{dt} = (p+1)t^p$

$$= \left[-\Gamma(p+1, u^{\frac{1}{p+1}}) \right]_e^\infty \quad dt = \frac{du}{(p+1)t^p}$$

$$= \left[-\Gamma(p+1, t) \right]_e^\infty \quad \because u = t^{p+1} \quad t = u^{\frac{1}{p+1}}$$

$$= \left[-\Gamma(p+1, \ln x) \right]_e^\infty \quad \because t = \ln(x)$$

$$= \Gamma(p+1, 1)$$

p should be > 0
and $p+1 \neq 0 \Rightarrow p \neq -1$